

# **GNN-based SHM for H3 Rocket Payload Fairing**

5-class defect detection + PINN regularisation + multitask centroid regression

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## **Problem setup**

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# SHM problem: two complementary analyses

**Structure:** CFRP / Al-honeycomb sandwich fairing (H3 Type-S, 1/6 section)    **Defects:** debonding / FOD / impact / delamination

## (A) Static FEM analysis

- Abaqus static stress field per defect scenario
- **10,897 nodes**  $\times$  **34 features** (disp, stress, temp, layup, . . . )
- Graph: FEM connectivity
- Task: **5-class node classification**  
+ defect centroid regression
- 700 simulations (560/140 split)
- Metric: OptF1 (threshold-optimal)

## (B) Dynamic guided-wave (GW) analysis

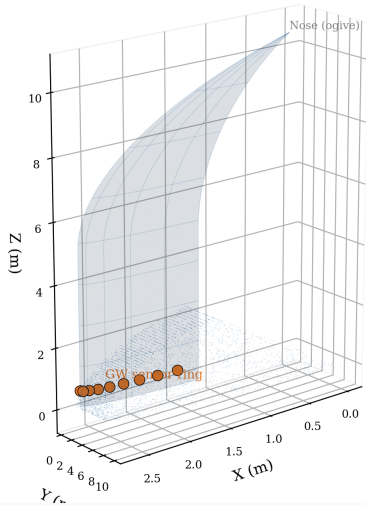
- Abaqus explicit GW propagation (piezo excitation)
- **9 sensors**  $\times$  **time series** (sampled at 4 MHz)
- Graph: sensor spatial network
- Task: **binary defect detection**  
+ wave-phase PINN (Level 3)
- Dataset: 108 samples (101 healthy / 7 defect, 9-sensor consistent)
- Model: MIFNO-GW (2-D factorised FNO, waveform-only)

## Node features (34 dims)

| Dims  | Feature group                                                    | Physical meaning    |
|-------|------------------------------------------------------------------|---------------------|
| 0–2   | Position ( $x, y, z$ )                                           | Node coordinates    |
| 3–5   | Normal ( $n_x, n_y, n_z$ )                                       | Surface orientation |
| 6–9   | Curvatures ( $k_1, k_2, H, K$ )                                  | Shell geometry      |
| 10–13 | Displacement ( $u_x, u_y, u_z, \ u\ $ )                          | Structural response |
| 14    | Temperature                                                      | Thermal loading     |
| 15–18 | Stress ( $s_{11}, s_{22}, s_{12}, \sigma_{\text{Mises}}$ )       | Failure indicators  |
| 19–23 | Principal stress, thermal $\sigma_M$ , log-strain                | Derived quantities  |
| 24–30 | Fibre direction, layup ( $0^\circ/45^\circ/-45^\circ/90^\circ$ ) | Anisotropy          |
| 31–33 | Circumferential angle, boundary/load flags                       | Topology            |

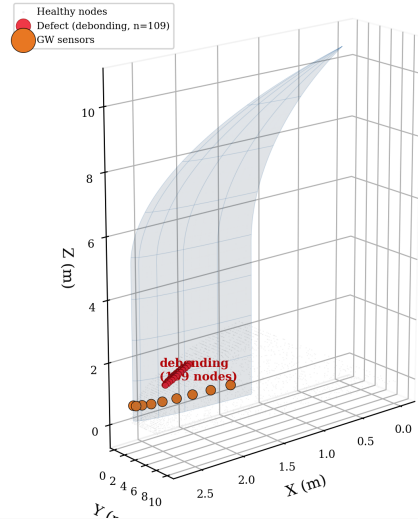
# H3 Fairing: 3D Structure & Defect Scenarios

**H3 Fairing — 1/6 Section**  
(CFRP/Al-honeycomb sandwich, 10,897 FEM nodes)



1/6 sector, 10,897 FEM nodes — CFRP/Al-honeycomb sandwich

**Ground-truth Defect Region**  
(debonding,  $z \approx 2386$  mm,  $r \approx 294$  mm)



Representative defect: **debonding**, 109 nodes

# Methods

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# Three modelling additions

## (A) Multi-task learning — defect centroid regression

Joint loss:  $\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}$

Shared backbone  $\rightarrow$  node classification head + graph-level MLP for  $(x, y, z)$  centroid.

GT centroid computed on-the-fly from defective nodes.

## (B) Level-1 PINN — stress gradient regularisation

High predicted defect probability should co-locate with high  $|\nabla \sigma_{\text{Mises}}|$ .

$$\mathcal{L}_{\text{stress}} = \frac{1}{N} \sum_i p_i^{(\text{def})} \exp(-|\Delta \sigma_i|/\sigma_0)$$

## (C) Level-2 PINN — static equilibrium residual $\nabla \cdot \sigma = 0$

Graph-edge finite differences on  $(s_{11}, s_{22}, s_{12})$ ; penalises defect predictions at nodes with low equilibrium violation.



**Best-performing architecture:** LocalGlobalAttentionGNN (inspired by LGSTA-GNN, Buildings 2025)

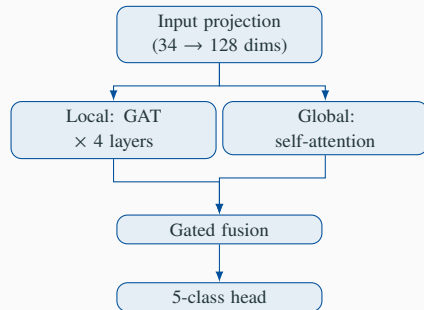
**Local branch** ( $\text{GAT} \times 4$  layers)

- Neighbourhood message-passing
- Captures local defect boundaries and stress concentrations

**Global branch** (Multi-head self-attention)

- Full  $N \times N$  attention over all 10,897 nodes
- Captures global mode shapes and structural coupling

**Fusion:** gated combination of both representations



Memory:  $O(N^2)$  global attention requires batch size = 1 on 24 GB GPU.

# Results

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## (A) Static FEM — architecture & PINN comparison (interim ep 40–100)

| Model             | Task     | OptF1        | AUC          | deb   | fod   | imp   | del   | Reg. RMSE     | NC = node |
|-------------------|----------|--------------|--------------|-------|-------|-------|-------|---------------|-----------|
| <b>LGSTA</b>      | NC+L1+L2 | <b>0.862</b> | 0.999        | 0.426 | 0.414 | 0.357 | 0.277 | —             |           |
| SAGE              | NC+L2    | 0.845        | 0.999        | 0.298 | 0.381 | 0.404 | 0.002 | —             |           |
| <b>LGSTA</b>      | MT+L1+L2 | 0.837        | 0.999        | 0.361 | 0.388 | 0.363 | 0.241 | <b>0.8 mm</b> |           |
| GATv2             | MT+L1+L2 | 0.829        | 0.999        | 0.342 | 0.315 | 0.359 | 0.000 | 0.7 mm        |           |
| SAGE              | MT+L1+L2 | 0.826        | 0.999        | 0.324 | 0.394 | 0.348 | 0.182 | 0.7 mm        |           |
| GCN               | MT+L1+L2 | 0.819        | 0.999        | 0.486 | 0.502 | 0.082 | 0.584 | 0.6 mm        |           |
| GIN               | MT+L1+L2 | 0.812        | 0.998        | 0.186 | 0.000 | 0.260 | 0.030 | 1.2 mm        |           |
| <b>Transolver</b> | MT+L1+L2 | <b>0.010</b> | <b>0.493</b> | —     | —     | —     | —     | —             |           |

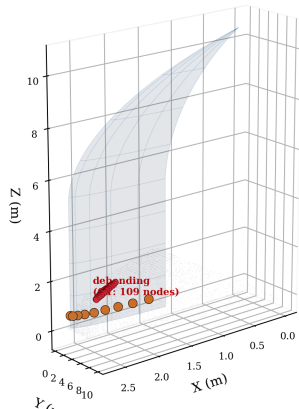
cls only    **MT** = +centroid regression    **L1** = stress-gradient PINN    **L2** = equil. residual PINN

Interim results (ep 40–100); all runs continue to 200 epochs.

# (A) Static FEM — Node-level Defect Localisation

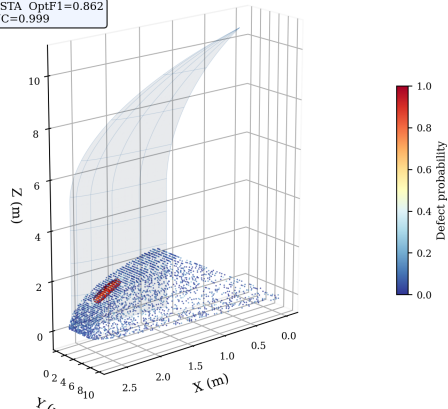
## H3 Fairing Node-level Defect Detection — Static FEM Analysis

(a) Ground-truth Defect Label



(b) LGSTA GNN Prediction (L1+L2 PINN)

LGSTA OptF1=0.862  
AUC=0.999



**Left:** ground-truth debonding labels (109 nodes, red). **Right:** LGSTA node defect probability.

Model concentrates high probability on the true defect region (OptF1 = 0.862, AUC = 0.999).

## (B) Dynamic GW — MIFNO-GW v2 (sensor-consistent)

**Model:** MIFNO-GW (waveform-only FNO)

2-D factorised FNO — Lehmann et al. 2025

**Input:** 9 sensors  $\times$  4,000 time steps

**Task:** binary defect / healthy classification

|         | Train | Val |
|---------|-------|-----|
| Samples | 86    | 22  |
| Healthy | 81    | 20  |
| Defect  | 5     | 2   |

**Training:** ep 200,  $\lambda_{\text{disp}} = 0.05$ ,

results pending (runs on Vancouver)

### Data quality fix (v1 $\rightarrow$ v2)

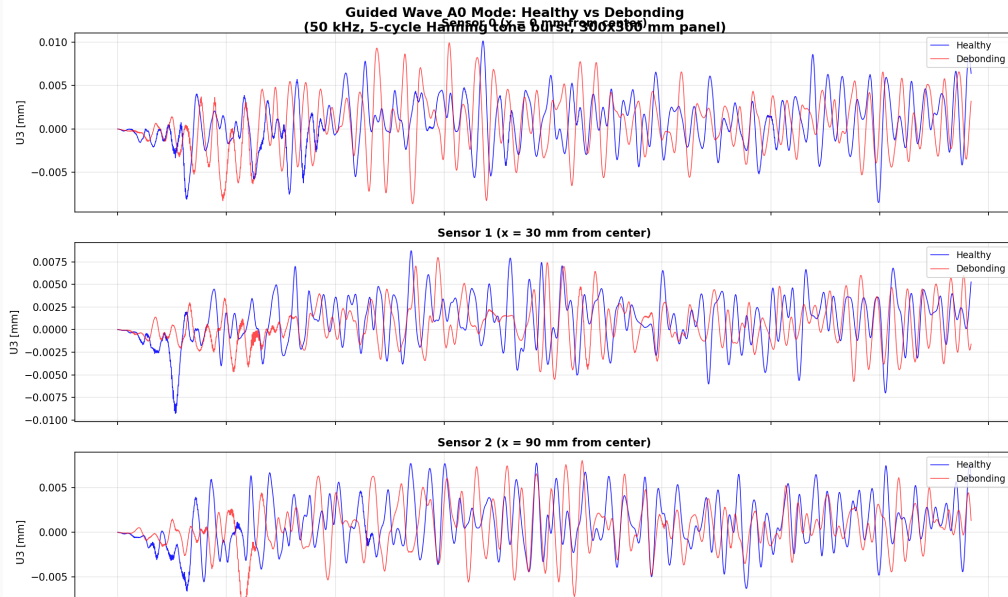
**Root cause found:** healthy CSVs (9 sensors, circumferential ring) and defect CSVs (99 sensors, axial line) used *different* physical sensor layouts. v1 F1 = 1.0 was a position-leakage artifact — the model learned to distinguish sensor-grid patterns, not wave signatures.

**Fix:** filter to 9-sensor position-consistent subset (108 samples); remove position branch from model (waveform-only FNO).

### Level-3 PINN — wave phase consistency

$$\mathcal{L}_{\text{disp}} = \left\langle w_{ij} \left[ \angle(\hat{u}_j \hat{u}_i^*) - \frac{\omega |dx|}{c} \right]^2 \right\rangle_{ij, \omega}$$

## (B) Dynamic GW — Wavefield: Healthy vs Debond



# Key findings

## (A-1) LGSTA wins at OptF1 = 0.862

Dual local-global attention: local GAT finds defect boundaries; global self-attention captures full structural coupling.

## (A-2) Multitask centroid regression: 0.6–1.2 mm RMSE

Shared backbone simultaneously classifies defect type *and* locates centroid — sub-mm accuracy on a 5 m structure.

## (A-3) L2 PINN alone (SAGE, 0.845) competitive with L1+L2

$\nabla \cdot \sigma = 0$  residual is the dominant physics term; stress-gradient L1 adds marginal gain.

## (A-4) Transolver collapses (AUC $\approx$ 0.49)

Designed for structured grids; fails on irregular FEM mesh even without physics loss.

## (B) GW dataset: position-leakage fixed; still small

v1 F1 = 1.0 was artifact (sensor layout  $\neq$  between healthy/defect). v2: 108 clean samples, waveform-only FNO. Need  $\gtrsim 50$  defect for L3 PINN

## Outlook

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## Next steps

- **200-epoch results** — GCN, GIN, LGSTA, GATv2, MeshGNN convergence (target: 2026-06-23)
- **GW dataset unification** — ONE Healthy Abaqus job (99-sensor) + augment  $\Rightarrow$  197-sample consistent dataset; enables L3 PINN & MIFNO-GW v3
- **Sim-to-real (OGW)** — Open Guided Waves omega-stringer dataset (18 GB, Zenodo 5105861): CFRP plate + omega stringer, 3 scenarios (intact / local / large debond), 50 kHz SLDV full wavefield. Run MIFNO-GW inference on virtual sensor traces extracted from 233 k scan points.
- **Cross-structure** — OOD: train fairing, test on stringer plate geometry
- **Fused static+dynamic** — concatenate LGSTA node logits with MIFNO-GW defect probability

### Target for WCCM meeting (2026-06-23)

Final results table with all 8 architectures at 200 epochs + ablation: NC vs. MT, L1 vs. L2 vs. L1+L2.